Semester 1 (Autumn)

Phase 1: Background and Project Design (September–October)

- Objectives:
  - Learn basic Y chromosome organization (PAR regions, X-degenerate regions, X-transposed regions, Ampliconic regions and Palindromes)
  - Understand sequencing coverage and depth.
  - Review existing genome browsers and visualization tools.
  - Define software requirements and user interface.

- Deliverables
  - Short literature review.
  - Specification of inputs, outputs, and desired visualizations.
  - Initial software design.

Phase 2: Data Processing and Prototype Development (November–December)

- Objectives
  - Obtain example coverage datasets (real or simulated).
  - Parse common genomic formats (BED, TSV, coverage tracks).
  - Implement basic visualization functions.

- Core features
  - Plot coverage along chromosome Y.
  - Display sequence-class annotations.
  - Compare multiple samples.

- Deliverables
  - First prototype capable of loading and visualizing coverage data.
  - Preliminary thesis Methods section.

Semester 2 (Spring)

Phase 3: Interactive Tool Development (February–March)

- Objectives
  - Develop the final application using Python (Plotly/Dash) or R (Shiny)
  - Add interactive functionality:
    - zooming
    - region selection
    - sample filtering
    - export of figures

- Deliverables: Fully functional software package.

Phase 4: Testing, Validation, and Thesis Writing (April–May)

- Objectives
  - Test the tool on multiple samples.
  - Evaluate visualization performance and usability.
  - Produce example biological case studies.

- Possible demonstrations:
  - Complete Y chromosome coverage.
  - Simulated partial deletions.

    - Regions with reduced coverage.

- Deliverables
    - Results chapter.
    - User documentation.
    - GitHub repository.

Phase 5: Thesis Finalization (May)

- Objectives
    - Complete thesis text.
    - Prepare defense presentation.
    - Final software polishing.

- Final outputs
    - Bachelor thesis.
    - Documented visualization software.
    - Public GitHub repository.
    - Example datasets and tutorial.